Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Understanding the XDR Kill Chain

The Stellar Cyber Open XDR Platform is organized around the **XDR Kill Chain**, a powerful framework for understanding and responding to threats on the security attack surface. The XDR Kill Chain identifies five separate stages on the attack landscape, each with its own associated tactics and techniques. Tactics and techniques include both those based on the MITRE | ATT&CK framework , as well as XDR-specific versions developed by Stellar Cyber.

Stellar Cyber developed the XDR Kill Chain to bring the MITRE | ATT&CK framework into the platform while addressing some of its shortcomings, such as a lack of high level stages, an inability to distinguish internal vs. external attacks, and a lack of a robust tagging mechanism for alerts. The XDR Kill Chain was designed to make analysts more efficient while enabling AI to better process alerts.

The XDR Kill Chain includes the following stages, each identified by its own distinctive color and icon. These colors and icons are used consistently throughout the user interface to reinforce stage identification.

XDR Kill Chain Stage	Icon	Summary	Associated MITRE | ATT&CK Tactics	Associated XDR Tactics	Classic Attacks
Initial Attempts		Attackers are trying to access your network.	Resource Development External Credential Access Reconnaissance Initial Access	XDR SBA External XDR NBA External XDR UBA	• Port scanning • External brute force login attempts • Phishing • Probes of known security holes
Persistent Foothold		Attackers are trying to maintain access to your systems in spite of defensive techniques.	Persistence Execution	XDR EBA External XDR Malw	• External trojans • Account manipulation

			Defense Evasion Command and Control	are XDR Intel	
Exploration		Attackers are inspecting your network, learning about your environment before taking further action.	Collection Discovery	Internal XDR NBA	• Internal port scans • External SQL dumpfiles • Suspicious SMB copies
Propagation		Attackers are trying to obtain additional privileges and access and control additional systems on your network.	Internal Credential Access Privilege Escalation Lateral Movement	Internal XDR UBA Internal XDR Malware	• Internal spyware • Internal trojans • Internal brute force
Exfiltration & Impact		Attackers are trying to steal data, potentially dumping it in a way that avoids detection, and/or vandalizing your systems and data.	Exfiltration Impact		• Syn floods • Ransomware • File action anomalies

Additional Tactics 6.0.0

MITRE ATT&CK has three tactics domains: Enterprise, Mobile, and Industrial Control Systems (ICS). In past releases, Stellar Cyber supported the Enterprise domain as described in the table above.

Starting in release 6.0.0, Stellar Cyber also supports the Mobile and ICS domains. Armis alert integration alerts contain ICS domains. See Integration of Third Party Native Alerts.

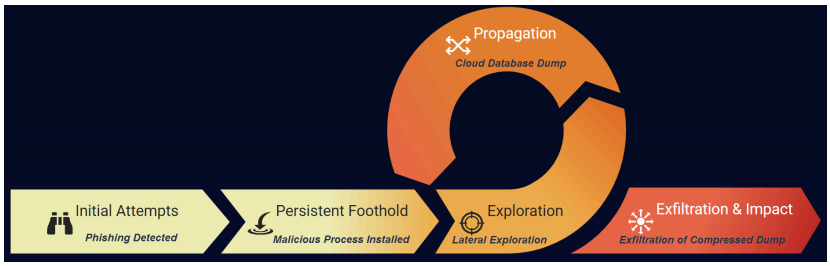
- For the Enterprise tactics, see Enterprise Tactics.
- For the Mobile tactics, see Mobile Tactics.
- For the ICS tactics, see ICS Tactics.

XDR Kill Chain Stage	Enterprise	Mobile	ICS
Initial Attempts	Reconnaissance TA0043 Resource Development TA0042 Initial Access TA0001 Credential Access TA0006	Initial Access TA0027 Credential Access TA0031	Initial Access TA0108
Persistent Foothold	Execution TA0002 Persistence TA0003 Defense Evasion TA0005	Execution TA0041 Persistence TA0028 Defense Evasion TA0030	Execution TA0104 Persistence TA0110 Evasion TA0103

XDR Kill Chain Stage	Enterprise	Mobile	ICS
	Command and Control TA0011	Command and Control TA0037	Command and Control TA0101
		Remote Service Effects TA0039	Impair Process Control TA0106
Exploration	Collection TA0009	Discovery TA0032	Discovery TA102
	Discovery TA0007	Collection TA0035	Collection TA0100
Propagation	Privilege Escalation TA0004	Privilege Escalation TA0029	Privilege Escalation TA0111
	Credential Access TA0006	Credential Access TA0031	Lateral Movement TA0109
	Lateral Movement TA0008	Lateral Movement TA0033	Inhibit Response Function TA0107
		Network Effects TA0038	
Exfiltration & Impact	Exfiltration TA0010	Exfiltration TA0036	Impact TA0105
	Impact TA0040	Impact TA0034	

XDR Kill Chain In Action

Consider the classic attack progression shown superimposed on the XDR Kill Chain, below, with a standard phishing attack moving through multiple stages to eventual data exfiltration:



- Alerts are triggered at each of the five XDR Kill Chain stages, identified by their associated tactic and technique, compatible with the MITRE | ATT&CK matrix.
- The XDR Kill Chain easily differentiates external alerts from internal/lateral ones and is able to capture complex attack progressions using its looped design between **Exploration** and **Propagation**.
- XDR Kill Chain stages are generally (but not strictly) progressive in nature. In other words, an alert *might* be related to a prior alert in a lower level on the XDR Kill Chain but sometimes is not.
- In addition to being assigned an XDR Kill Chain stage, alerts are also scored for severity and fidelity, helping you focus on the alerts that matter. Stellar Cyber also automatically correlates related alerts and groups them into cases.

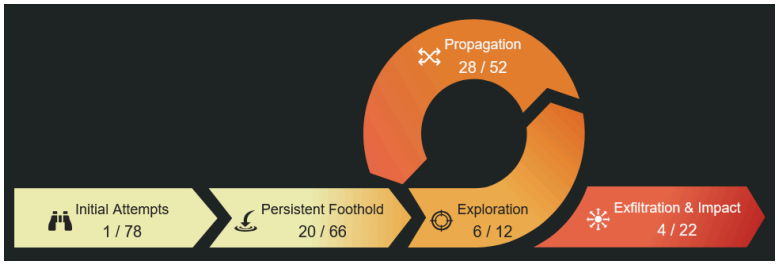
About XDR Tactics

The XDR Kill Chain augments MITRE | ATT&CK-based tactics and techniques with its own tactics to cover threat intelligence, malware detection, and anomaly detections. Threat intelligence is based on reputation analysis, malware detection on known signatures, and anomalies on deviations from standard observed behavior patterns:

- **XDR Malware** – Alerts for malware.
- **XDR Intel** – Alerts for threat intelligence.
- **XDR User Behavior Analytics (UBA)** – Alerts for user anomalies.
- **XDR Network Behavior Analytics (NBA)** – Alerts for network anomalies.
- **XDR Endpoint Behavior Analytics (EBA)** – Alerts for host-based anomalies.
- **XDR Sensor Behavior Analytics (SBA)** – Alerts for operational and injection-based anomalies.

XDR Kill Chain in Stellar Cyber

The XDR Kill Chain appears throughout Stellar Cyber, most prominently as the centerpiece of the Home Page dashboard, giving you a bird's-eye summary of critical and total alerts at each Kill Chain stage in the current time interval:



You'll also find the XDR Kill Chain's familiar stages and icons in the Cases page, the various Alert pages, Threat Hunting, and Reporting, among others.

Anatomy of an Alert

Stellar Cyber categorizes alerts according to their XDR Kill Chain stage, tactic, technique, and alert name:

Stellar Cyber generates alerts by using machine-learning algorithms to search for specific security-related events.

The table below summarizes these relationships for **IP/Port Scan Anomaly** detection:

XDR Kill Chain Stage	Initial Intrusions
Tactic (MITRE | ATT&CK or XDR-Extended)	TA0043: Reconnaissance
Technique	T1595: Active Scanning
Alert	External IP/Port Scan Anomaly on 135.x.y.25

Other Components of the Stellar Cyber Data Model

Alerts are just one part of the Stellar Cyber Data Model. The Data model represents all data in its different forms within Stellar Cyber and is useful to keep in mind when investigating alerts or cases:

- **Raw Event** - Stellar Cyber starts with raw data ingested from security and IT tools, including Stellar Cyber sensors.
- **Signal** - Stellar Cyber identifies critical Raw Events with special security value as signals. Signals are stored in their own dedicated index and are preserved longer than

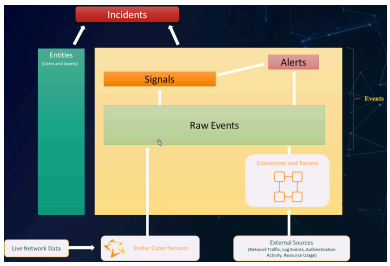
raw events because of their usefulness (for example, domain creations or Windows Defender events).

- **Alert** - A specific security-related event generated by Stellar Cyber using machine-learning algorithms.

  Together, raw events, signals, and alerts are referred to generically as *events*.

- **Entity** - Stellar Cyber considers assets or users within the enterprise as entities.
- **Case** - Multiple alerts and entities correlated together into a unified security threat.

The figure below illustrates the relationships between the components of the Stellar Cyber data model.



XDR Kill Chain Stages Described

This section describes the XDR Kill Chain stages in detail.

Initial Attempts

Attackers are looking for a way into the network or the hosts connected to the network. Typical techniques include:

- External port scanning.
- Brute force login attempts.
- Phishing.
- Probes of known security holes.

During the Initial Attempts stage, attackers perform reconnaissance, inspect the network for initial footholds, attempt to establish resources for further operations, and, crucially, attempt to steal credentials that can enable access, perhaps via brute force or credential dumping.

Persistent Foothold

Attackers are trying to ensure their ongoing access to network resources, regardless of defensive measures such as system reboots or credential updates. Typical techniques include:

- Copy operation (`rcp`, `rsync`, etc.) via SSH after account access is established.
- Malware delivery by email, removable memory device, or HTTP transfer using tools such as `wget` or `curl`.
- Removing or disabling security software, perhaps via scripts.
- External trojans.

This phase also involves the owner of the malware establishing an illicit communication channel to the infected system. Nearly any protocol can be used and is most often selected to appear as normal traffic as much as possible so that it is allowed by any firewalls or routers in the path.

For systems that do not have a public IP address, the communication link is typically initiated by the infected system.

Exploration

Attackers are inspecting your network, performing reconnaissance to learn about your environment before taking further action. Typical techniques include:

- Use of native OS tools for discovery of accessible network shares, browser bookmarks, and peripherals.
- Network sniffing, screenshots, and keyboard logging to collect audio, video, and email.
- Internal IP/port scanning, suspicious file copies, URL reconnaissance, and SQL dumpfiles.

Stellar Cyber looks for deviations from observed network behavior analysis baselines at this stage, preemptively alerting you to ongoing exploration of your resources by attackers.

Propagation

Attackers are attempting to escalate their privileges, spreading their access to other areas of your network via lateral movement, malware, and internal credential access. Typical techniques include:

- Internal trojans, spyware, and brute force attacks.
- Internal credential stuffing.
- Remote desktop exploits.

Stellar Cyber looks for deviations from observed user behavior analysis baselines at this stage, as well as common credential access, remote desktop, and SMB exploits.

Exfiltration and Impact

When sufficient control is established, attackers use infected systems as desired, either by direct access or by pre-programmed time release. Typical techniques include:

- Leaks of PII
- Data exfiltration through remote desktop, Office 365, or DNS tunnels.
- Cryptojacking

This is the stage during which intended damage is actually done, including:

- Collecting and transmitting sensitive user information
- Triggering ransomware
- Destroying systems
- Modifying data

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